COMPUTER NETWORKS AND DATA COMMUNICATION

[CPE2206]

Bachelor of Engineering in Electronics and Computer Engineering (BENG ECE)

Year II, Semester II

Academic Year 2025/2026

PREREQUISITES

- Introductory course

COURSE DESCRIPTION:

- This course will introduce students to computer networks and data communications.
- The course addresses the concepts of computer networks and how they are applied in the day-to-day work of organizations, data communications highlighting how data and communications relate with signals during transmission within the domain of computer networks.

COURSE OBJECTIVES

- Introduce students to computer networks, data communication concepts and their application in organizations.
- Enable students appreciate the different concepts in different communication models such as OSI and other models.
- Introduce students to different forms of data and signal transmission.
- Learn the difference between the OSI model, TCP/IP architectural model and the three-layer model.

COURSE LEARNING OUTCOMES

- ❑ By the end of the Course, the student will be able to do the following:
    - ✓ Define and differentiate between computer networks and data communication

- ✓ Explain the different forms of media used in communications, outline their advantage and disadvantages over the other.
- ✓ Illustrate computer networking skills and explain the modes of data and media transmissions.
- ✓ Master the terminology and concepts of the OSI reference model and the TCP/IP reference model;
- ✓ Master the concepts of protocols, network interfaces, and design/performance issues in local area networks and wide area networks.

MODE OF DELIVERY

- Lectures
- Practical
- Discussions
- Q & A

MODE OF ASSESSMENT

- Tests/Assignments/Practical: 40%
- Written Examination: 60%

ASSESSMENT SCHEDULES

- Course Work Test 1: Week 6 {Friday 6th March 2026, 10am}
- Course Work Test 2: Week 10 {Friday 3rd April 2026, 10am}

LECTURE 1

INTRODUCTION TO DATA COMMUNICATION AND NETWORKS

INTRODUCTION TO DATA COMMUNICATIONS

The term communication can be defined as the process of transferring messages between entities.

When we communicate, we are sharing information. This sharing can be local or remote. Between individuals, local communication usually occurs face to face, while remote communication takes place over distance.

The term ***telecommunication,*** which includes telephony, telegraphy, and television, means communication at a distance *(tele* is Greek for "far").

The word ***data*** **r**efers to information presented in whatever form is agreed upon by the parties creating and using the data.

Data communications: *is the process of transferring data from one place to another, and vice versa using computer and communication technologies.*

It can also be defined as the exchange of data between two devices via some form of transmission medium such as a wire cable.

Based on what these two entities are, there are three basic types of communication:

- *Human to Human*
- *Computer to Computer*
- *Human to Computer*

For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

GENERAL COMMUNICATION MODEL

CHARACTERISTICS OF DATA COMMUCATIONS SYSTEMS

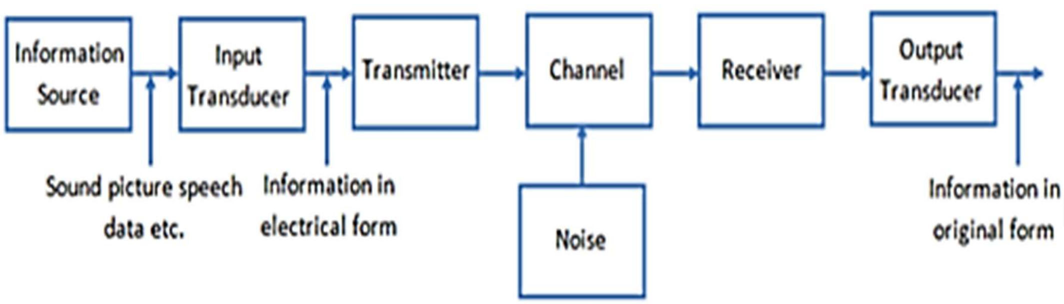
The effectiveness of a data communications system depends on four fundamental characteristics: *delivery, accuracy, timeliness*, and *jitter*.

Delivery: The system must send data to the correct destination so that only the intended device or user receives it.
Accuracy: The system must deliver data correctly; any data altered during transmission and not corrected is unusable.

Timeliness: The system must deliver data on time; late data is useless. For audio and video, data must be delivered in order, as produced, and without significant delay—this is called real-time transmission**.**

Jitter: Jitter is the inconsistent arrival time of data packets. Even if packets are sent at regular intervals, they may arrive at the receiver at uneven times. This variation causes audio to sound choppy or video to appear jerky, especially in real-time applications like video calls and streaming, resulting in reduced quality.

BLOCK DIAGRAM OF COMMUNICATION SYSTEM

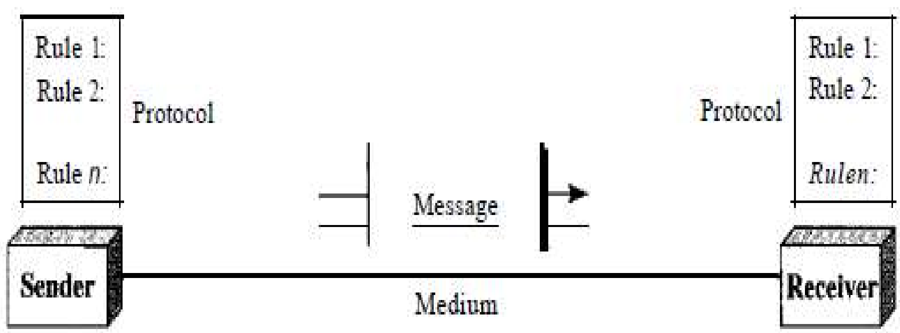


- **Information Source:** Generates the msg (voice, data, image, video).
- **Input Transducer:** Converts physical signals into electrical signals (E.g., microphone: sound -> electrical).
- **Transmitter:** Processes the signal for transmission. E.g., encoding, modulation, amplification.
- **Communication channel:** Medium through which signals travel **(**E.g., air, cable, optical fibre). Noise -> distortion.
- **Receiver:** Recovers the original signal. (E.g., demodulation, decoding, filtering)
- **Output Transducer:** Converts electrical signal back to physical form (E.g., speaker: electrical->sound)
- **Destination:** Final user or device that receives the information.

COMPONENTS OF DATA COMMUNICATIONS

A data communications system has five components:

1. **Message.** The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender.** The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver.** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. **Transmission medium.** The transmission medium is the path by which a message travels from sender to receiver. Some examples of transmission media
5. **Protocol.** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.



DATA REPRESENTATION TECHNIQUES

Information today comes in different forms such as text, numbers, images, audio, and video.

Text
Text in data communication is represented using bits (0s and 1s)**.** Characters are assigned bit patterns through a process called coding**.** Unicode is the most common coding system and represents characters from all languages, usually using 32 bits per character**.**

Numbers
Numbers are represented using bit patterns as well. Instead of using codes like ASCII, numbers are directly converted into binary form. This makes mathematical calculations easier and faster for computers.

Images
Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the *resolution.*

Audio
Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete.
Video
A video is a recording or broadcast of moving pictures. It can be continuous**,** like live TV, or made of a series of images shown in sequence to create motion.

DATA COMMUNICATION SIGNALS

Two communication signals:

Analog Signal: refers to continuous signals that vary smoothly over time and represent data using variations in amplitude, frequency, or phase. Example: Voice transmitted over a telephone line, radio signals.

Digital Signal: Discrete signals with specific levels, typically represented as binary (0s and 1s). Example: Data from computers, digital audio

MODEM

Short for modulator/demodulator (Mo/dem. A communications device that converts one form of a signal to another.

- Modulation. Converting digital signals into analog signals.
- Demodulation. Converting analog signals back into digital signals.

MODES OF DATA TRANSMISSION

When data are transmitted from one point to another, *four modes* of transmission can be identified:

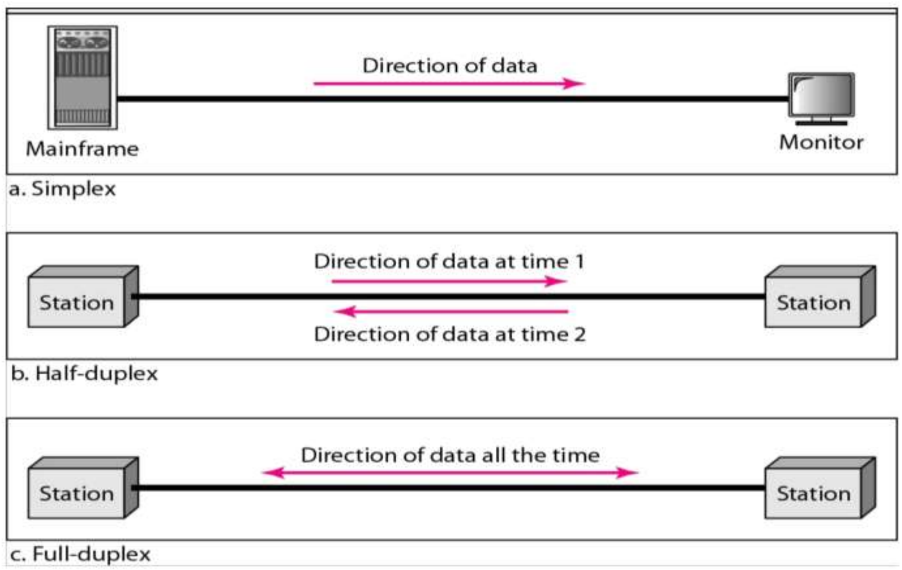
- Simplex (SX) mode
- Half Duplex (HDX) mode
- Full Duplex/Duplex (FDX/DX) mode
- Full/Full Duplex (F/FDX) mode

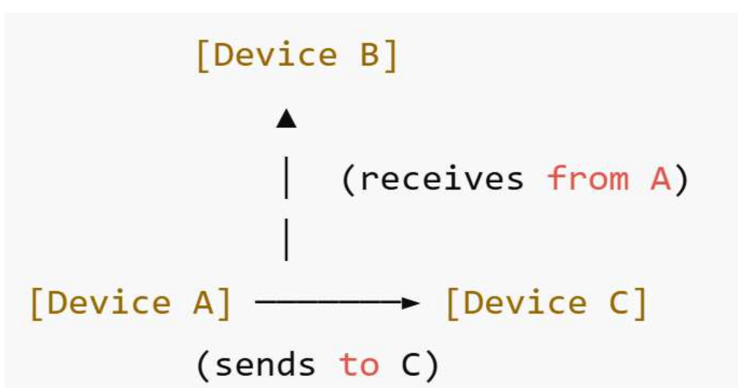
Simplex (SX) mode allows communication in only one direction. One device transmits, the other receives. There is no feedback from the receiver. Example: Radio, TV broadcasting. Also called **one-way, transmit-only, or receive-only** communication.

Half Duplex (HDX) mode allows both devices to transmit and receive, but not at the same time**.** When one device is sending, the other is receiving.
The full channel capacity is used one direction at a time**.** Example: Two-way radio (radio call/ walkie talkie)**.**

Full Duplex (FDX) mode allows both devices to transmit and receive at the same time**.** It is used when continuous two-way communication is needed.
The channel's capacity is shared between both directions. Example: Telephone calls or computers sending and receiving data simultaneously.

Full/Full Duplex (F/FDX) mode allows simultaneous two-way communication**,** but not between the same two devices**.**
One device can send to a second device while receiving from a third device**.**
It works only on **multipoint circuits**.
Example: **Postal system** — sending a letter to one person while receiving a letter from another.





d. Full/full Duplex

REASONS FOR STUDYING DATA COMMUNICATIONS

- To build a strong foundation in networking, digital technology, communication methods, etc.
- Data communications is a critical skill for data scientists, who need to share their insights and recommendations with different audiences like clients, managers, colleagues and the general public
- To develop skills and knowledge essential to design, build, maintain and manage networks and communication systems in any organization.
- Data communication and networking is a truly global area of study because the technologies enable global communication.
- Data communication is something we experience in our every-day-lives, and a good understanding of it improves the way we use this technology.

INTRODUCTION TO COMPUTER NETWORKS

A **network** is a set of devices (often referred to as nodes*)* connected by communication links.

A **node** can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

A **Computer network** is a system in which a number of independent (autonomous) computers are linked together to share data and peripherals, such as files and printers.

In the modern world, computer networks have become almost indispensable. All major businesses and governmental and educational institutions make use of computer networks to such an extent that it is now difficult to imagine a world without them.

Two computers are said to be interconnected if they are able to exchange information.

BENEFITS OF COMPUTER NETWORKS

Following are the benefits of a computer network:

- **Resource sharing:** Computer network help users to share resources, like: printer, scanner, storage devices (hard disk drive), internet etc. It also enables the users to share data files and software.

- **User Communication**: Network allows users to communicate using e-mail, newsgroup, video conferencing, messenger, Viber, WhatsApp etc.

- **Increases span of Management:** A single manager can monitor and supervise large number of subordinates with the help of information system through computer network

- **Reduces gap between Departments:** Inter department communication becomes easier if different departments are connected through computer network.

- **Voice Over IP (VoIP):** Voice over internet protocol (VoIP) allows users to communicate using standard Internet Protocol (IP) rather than tradition telephone line.

NETWORK CRITERIA

In computer networks, a good network is usually judged by three core criteria which include:

- *Performance*
- *Reliability*
- *Security*

❑ Performance:

This refers to how fast and efficiently the network operates.

Factors that determine network performance include:

- **Throughput** – the amount of data delivered per unit time.
- **Latency (delay)** – how long data takes to travel from source to destination.
- **Response time** – time between a request and the response.
- **Bandwidth** – maximum data-carrying capacity of the network.

Network performance is affected by the following factors:

- Number of users.
- Type of transmission medium.
- Hardware capabilities (routers, switches, servers).
- Software and protocols in use.

❑ Reliability:

Reliability measures how dependable the network is.

It includes:

- **Frequency of failure** – how often the network goes down
- **Recovery time** – how quickly the network is restored after failure
- **Fault tolerance** – ability to continue operating when part of the network fails

A reliable network ensures continuous availability and minimal disruption of services.

❑ Security:

Security protects data and network resources from unauthorized access and attacks and having policies and procedures to recover from breaches or data loss.

It involves:

- **Confidentiality** – only authorized users can access data.
- **Integrity** – data is not altered during transmission.
- **Availability** – services remain accessible to legitimate users.

Security mechanisms include:

- Authentication and authorization
- Encryption
- Firewalls and intrusion detection systems

PHYSICAL STRUCTURES

TYPES OF CONNECTIONS:

A network consists of two or more devices connected through **links**. A **link** is a communication pathway that transfers data from one device to another.

There are **two main types of connections**:

1. **Point-to-Point Connection**
2. **Multipoint (Multi-drop) Connection**

1. Point-to-Point Connection

A point-to-point connection is a direct link between two devices only. The entire capacity of the link is reserved for communication between these two devices. It may use cables or wires, microwave transmission or Satellite links.

Example:
Using a TV remote control to change channels is a point-to-point connection because the signal is sent directly from the remote to the TV.

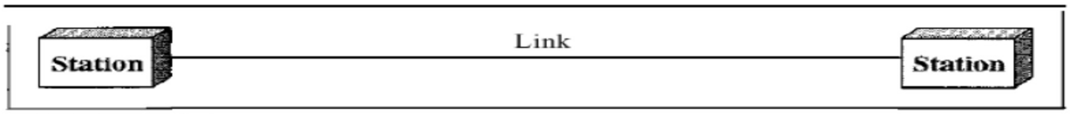
2. Multi-point (Multi-drop) Connection

A multipoint connection (also called a multi-drop connection) is one in which more than two devices share a single communication link.

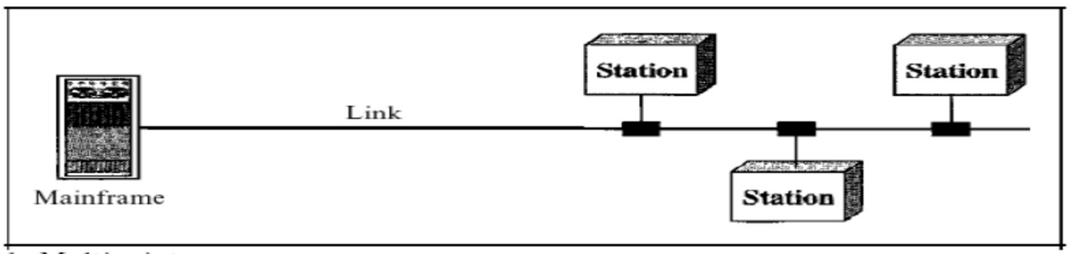
In a multipoint environment, the capacity of the channel is shared, either:

- **Spatially shared** – multiple devices transmit at the same time
- **Time-shared** – devices take turns using the link

Example:
Computers connected to a single bus cable in a LAN.



a. Point-to-point



b. Multipoint